Update on Project 7: Security

Cody Gunton - July 8, 2026 https://codygunton.github.io/talks-and-writing/2026-07-08-zkevm-breakout/

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RISC-V Compliance Testing

https://github.com/eth-act/zkevm-test-monitor/https://eth-act.github.io/zkevm-test-monitor/
Upgraded RISC-V test suites to ACT v4.0.0; exposed more bugs

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Formal RISC-V Compliance of ZisK

https://github.com/eth-act/zisk-fv
Gap between "AI wrote me some Lean proofs" and "I am convinced this is" is huge.
Progress is mostly technical; summary is here for those interested.
Met with Héctor Masip of ZisK, who's now ramping up to review and contribute.

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RISC-V Fuzzing

Near-final deliverables are a fuzzing grant; code released at the end of the month. Built on Ere
Rust and pure RISC-V
Don't want to front-run too much :)

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Innovations and ArkLib

Flock proof system: no more algebraic hashes .
ArkLib#449: Perfect completness of sumcheck; useful for WHIR, Flock and more. ArkLib#516: Ring switching formalism; will be useful for formalizing Flock.

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https://github.com/frisitano/evm-sail

Experiment with a formal specification in Sail machine spec language
Same language used for official RISC-V specs.
sail-to-lean backend already in use in zisk-fv and its ancestors.
Started June 13; sail-to-C compilation is passing ALL EEST TESTS .
My dream: feature parity with python specs, then replace them.

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Thanks for your attention!

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